

Specializing Foundation Models via Mixture of Low-Rank Experts for Comprehensive Head CT Analysis

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Abstract. Foundation models pre-trained on large-scale datasets demonstrate strong transfer learning capabilities; however, their adaptation to complex multi-label diagnostic tasks—such as comprehensive head CT finding detection—remains understudied. Standard parameter-efficient fine-tuning methods such as LoRA apply uniform adaptations across pathology types, which may limit performance for diverse medical findings. We propose a Mixture of Low-Rank Experts (MoLRE) framework that extends LoRA with multiple specialized low-rank adapters and unsupervised soft routing. This approach enables conditional feature adaptation with less than 0.5% additional parameters and without explicit pathology supervision. We present a comprehensive benchmark of MoLRE across six state-of-the-art medical imaging foundation models spanning 2D and 3D architectures, general-domain, medical-domain, and head CT-specific pretraining, and model sizes ranging from 7M to 431M parameters. Using over 70,000 non-contrast head CT scans with 74 annotated findings—including hemorrhage, infarction, trauma, mass lesions, structural abnormalities, and chronic changes—our experiments demonstrate consistent performance improvements across all models. Gains vary substantially: general-purpose and medical-domain models show the largest AUC improvements (DINOv3-Base: +4.7%; MedGemma: +4.4%), whereas 3D CT-specialized or domain-specific models show more modest gains (+0.2–1.2%). The combination of MoLRE and MedGemma achieves the highest average detection AUC of 0.917. These findings highlight the importance of systematic benchmarking on target clinical tasks, as pretraining domain, architecture, and model scale interact in non-obvious ways.

Keywords: Foundation Model · Head CT · Mixture of Experts.

1 Introduction

Foundation models pre-trained on massive datasets have fundamentally shifted the paradigm of medical image analysis, demonstrating strong zero-shot and few-shot transfer capabilities across a wide range of tasks [23, 17, 4, 10, 32, 1, 16, 15]. However, while these models achieve impressive performance on standardized benchmarks, their effectiveness in complex multi-label clinical scenarios—such as comprehensive non-contrast head CT analysis involving dozens of heterogeneous findings—remains insufficiently understood. Existing work largely focuses on single-organ detection or narrow subsets of abnormalities [28, 8], leaving open questions about how foundation models handle the high-dimensional diagnostic complexity underlying automated radiology report generation [24, 11, 30].

Adapting large foundation models to such specialized clinical tasks typically relies on parameter-efficient fine-tuning (PEFT) [5], with Low-Rank Adaptation (LoRA) [12, 20] emerging as a dominant approach. LoRA approximates weight updates through low-rank decomposition, enabling efficient adaptation with minimal additional parameters. However, standard LoRA applies a uniform low-rank update across all inputs, implicitly assuming that a single adaptation suffices for all pathology types. In comprehensive head CT analysis, this uniformity can lead to knowledge interference [25], where features required to detect acute hemorrhage, chronic ischemia, trauma, and subtle structural abnormalities must compete for the same limited adaptation capacity.

Recent advances in mixture-of-experts architectures enable conditional, input-dependent specialization by activating different subnetworks for different inputs [14, 26, 7, 18]. Motivated by these principles, we propose *Mixture of Low-Rank Experts (MoLRE)*, a parameter-efficient extension of LoRA that enables conditional adaptation via unsupervised soft routing with less than 0.5% additional parameters. Several recent works have explored combining mixture-of-experts mechanisms with low-rank adaptation, including MoLE [29], LoRAMoE [6], and MoELoRA [22]. These methods are closely related in that they use multiple low-rank adapters to enable conditional parameter-efficient adaptation. However, they have primarily been developed for natural language processing settings, where routing is typically performed over token-level representations and often involves hard or semi-hard expert selection. In contrast, MoLRE targets volumetric medical imaging, uses unsupervised soft routing conditioned on image embeddings, and is systematically benchmarked across heterogeneous 2D and 3D foundation models for large-scale multi-label head CT analysis.

Our main contributions are as follows:

- We adapt mixture-based low-rank specialization to comprehensive head CT analysis through MoLRE, which combines multiple low-rank experts with unsupervised soft routing for volumetric medical imaging.
- We integrate MoLRE into both 2D slice-based and 3D volumetric foundation model pipelines, enabling conditional adaptation at slice-level or volume-level representations depending on the model architecture.
- We present a large-scale benchmark across six heterogeneous 2D and 3D medical imaging foundation models, spanning general-domain, medical-domain,

and head CT-specific pretraining, on more than 70,000 non-contrast head CT scans with 74 neurological findings.

- We empirically show that the benefit of conditional low-rank specialization depends on the interaction between pretraining domain, model architecture, and model scale, with larger gains for general-purpose and medical-general foundation models and smaller but non-zero gains for task-specific volumetric models.

2 Method

2.1 Mixture of Low-Rank Experts (MoLRE)

MoLRE extends LoRA by introducing K specialized low-rank experts with learned input-dependent routing. Given an input feature $\mathbf{x} \in \mathbb{R}^d$, the adapted output is computed as:

$$\mathbf{h} = \mathbf{W}_0 \mathbf{x} + \sum_{i=1}^K g_i(\mathbf{x}) \cdot \Delta \mathbf{W}_i \mathbf{x} \quad (1)$$

where $\mathbf{W}_0 \in \mathbb{R}^{d_{out} \times d}$ is the frozen pre-trained weight matrix, and $\Delta \mathbf{W}_i = \mathbf{B}_i \mathbf{A}_i$ represents the i -th expert’s low-rank adaptation with $\mathbf{A}_i \in \mathbb{R}^{r \times d}$ and $\mathbf{B}_i \in \mathbb{R}^{d_{out} \times r}$, where $r \ll \min(d, d_{out})$ is the rank. The router network $g(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^K$ computes expert mixing weights via a two-layer MLP with softmax normalization:

$$\mathbf{g}(\mathbf{x}) = \text{Softmax}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) \quad (2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_h \times d}$, $\mathbf{b}_1 \in \mathbb{R}^{d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{K \times d_h}$, and $\mathbf{b}_2 \in \mathbb{R}^K$, with d_h denoting the router hidden dimension. The softmax ensures $\sum_{i=1}^K g_i(\mathbf{x}) = 1$ and $g_i(\mathbf{x}) \geq 0$, enabling all experts to contribute with varying degrees based on the input.

2.2 Integration with Foundation Models

For 2D foundation models (e.g., DINOv3 [27], MedGemma [17], MedImageInsight [4]), we process volumetric medical scans with S slices using slice-level feature extraction followed by MoLRE adaptation and attention-weighted pooling [13, 20, 3]. The router network $g(\mathbf{x})$ operates on individual slice features, enabling slice-specific expert selection, which is particularly beneficial for head CT where pathologies may be spatially localized. The complete inference pipeline for 2D foundation models is summarized in Algorithm 1.

For 3D models (e.g., Pillar0-HeadCT [1]) that natively process volumetric inputs, MoLRE is applied to spatially pooled volumetric features, with expert routing based on holistic volume-level representations rather than individual slices. The MoLRE formulation is identical across 2D and 3D models, differing only in the specialization level (slice-level for 2D, volume-level for 3D). Figure 1

Algorithm 1 MoLRE for 2D Vision Foundation Models

Require: Volumetric medical scan $\mathbf{X} \in \mathbb{R}^{B \times M \times S \times H \times W}$
Require: Foundation model f_θ , MoLRE module MoLRE_ϕ

- 1: Reshape: $\mathbf{X}' \leftarrow \text{reshape}(\mathbf{X}, [B \cdot S, M, H, W])$
- 2: Extract features: $\mathbf{F} \leftarrow f_\theta(\mathbf{X}') \{[B \cdot S, d]\}$
- 3: Apply MoLRE: $\mathbf{F}' \leftarrow \text{MoLRE}_\phi(\mathbf{F}) \{[B \cdot S, d]\}$
- 4: Reshape: $\mathbf{F}' \leftarrow \text{reshape}(\mathbf{F}', [B, S, d])$
- 5: **Attention pooling:**
- 6: $\mathbf{Q} \leftarrow \text{expand}(\mathbf{q}_{\text{learnable}}, B) \{[B, 1, d]\}$
- 7: $\alpha \leftarrow \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{F}'^T}{\sqrt{d}}\right) \{[B, 1, S]\}$
- 8: $\mathbf{h} \leftarrow \alpha\mathbf{F}' \{[B, d]\}$
- 9: Classify: $\mathbf{y} \leftarrow \text{Sigmoid}(\mathbf{W}_c\mathbf{h}) \{[B, C]\}$
- 10: **return** Multi-label predictions $\mathbf{y}$

illustrates the 2D pipeline; for 3D models, MoLRE operates directly on the pooled volumetric embedding.

All models were trained end-to-end using a unified protocol. For 2D foundation models, we used multi-label focal loss [19] with $\gamma = 2.0$ and prevalence-based class weights α_t , and applied parameter-efficient fine-tuning with LoRA (rank $r = 8$, scaling $\alpha = 16$) extended by MoLRE with $K = 6$ low-rank experts (rank $r_e = 8$) and router hidden dimension $d_h = 256$. Optimization used AdamW [21] with learning rates of 10^{-3} for classification heads and 10^{-4} for LoRA/MoLRE parameters, alongside Repeat Factor Sampling [9] to address rare findings. MoLRE routing was learned fully unsupervised via the task loss. For 3D foundation models, we followed the published training protocols. Models were trained for at least 20 epochs with early stopping after 5 epochs of no validation AUC improvement; results are reported from the best validation checkpoint.

On DINOv3-Base, MoLRE was robust to expert number and rank, with mean AUC remaining 0.902–0.903 for $K \in \{3, 6, 9\}$ and 0.901–0.902 for $r_e \in \{4, 8, 12, 16\}$; we therefore used $K = 6$ and $r_e = 8$ as a performance/overhead trade-off. Applying MoLRE only to the last 3 or 6 layers reduced mean AUC to 0.889 and 0.897, supporting all-layer insertion. On an A100 40GB GPU with batch size 1, DINOv3-Base+MoLRE required < 100 ms per preprocessed scan and 0.58 GB peak GPU memory.

3 Results and Discussion

3.1 Dataset and Preprocessing

Anonymized non-contrast head CT (NCCT) volumes were retrospectively collected from nine centers across different geographical regions, with ethics approvals waiving informed consent. Scans were acquired on Siemens Healthineers, GE HealthCare, and Canon Medical Systems scanners. Exclusion criteria included age under 18 or missing axial reconstructions. In total, 72,756 NCCT studies met inclusion criteria, of which 65,542 were used for model development

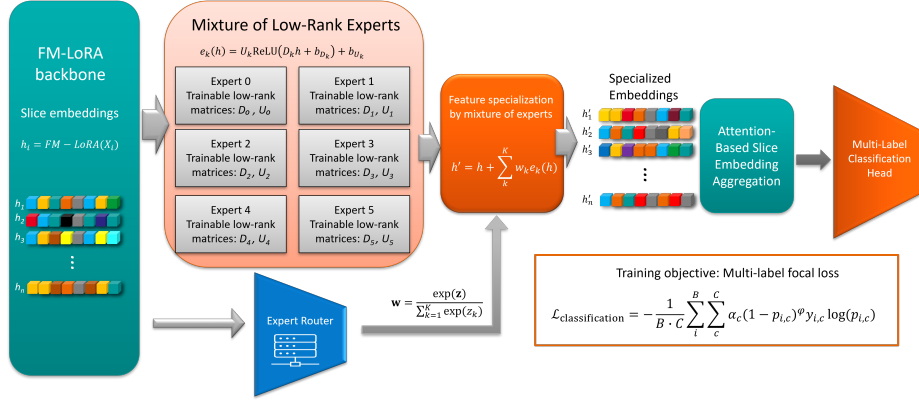


Fig. 1. MoLRE: Parameter-efficient specialization for foundation models. Illustrated for 2D models (e.g., DINOv3), MoLRE employs (1) multiple low-rank expert adapters for feature transformations, (2) an unsupervised soft router that learns to weight experts based on input features, and (3) attention-weighted pooling to aggregate slice-level features into volume-level representations. For 3D models (e.g., Pillar0-HeadCT), MoLRE is applied to spatially-pooled volumetric features, enabling conditional adaptation without explicit pathology supervision.

(58,251 for training and 7,291 for system optimization) and 7,214 were held out for independent evaluation. All splits were performed at the patient level, ensuring that scans from the same patient—including repeat and follow-up studies—were assigned exclusively to a single partition to prevent data leakage across training, optimization, and evaluation sets.

Leveraging recent advances in LLM-based medical content generation [2], we automatically generated labels for 74 neurological findings from radiology reports using a private GPT-4-mini model deployed on a secure network. Finding categories were predefined by a neuroradiologist (Figure 3), encompassing hemorrhage, vascular, trauma, mass/lesion, structural, and chronic/other categories. To improve labeling quality, prompts were split into batches of 15 findings and restricted to the impression section to reduce false positives in follow-up studies. Label accuracy was validated on 618 cases annotated by a neuroradiologist, achieving an average of 1.10 false positive findings and 0.16 false negative findings per case, with an overall finding-level accuracy of 0.982.

Preprocessing involved automatic alignment of axial NCCT volumes to a standard reference frame, resampling to a 1-mm in-plane and 4-mm out-of-plane resolution, and normalization using HU windows: 0–80 HU, -20–180 HU, and -800–2000 HU, scaled to 0–1. Spatial augmentations include elastic deformation ($\alpha \in [0, 200]$, $\sigma \in [10, 13]$), small rotations ($\theta_x, \theta_y, \theta_z \in [-0.1, 0.1]$ radians), and anisotropic scaling (scale factor $\in [0.85, 1.15]$). Intensity augmentations consist of multiplicative brightness modulation ($m \in [0.99, 1.01]$) and additive Gaussian noise ($\mathcal{N}(0, \sigma^2)$ with $\sigma^2 \in [0, 0.03]$). Random axis-wise mirroring is applied with probability $p = 0.5$ for each axis independently.

Table 1. Multi-label head CT detection performance and model complexity. Mean AUC $\pm$ std across 74 neurological findings and parameter counts for each foundation model. All 2D models use LoRA fine-tuning with attention-based slice aggregation. Pillar0-HeadCT is fully fine-tuned. DeepCNTD is excluded from MoLRE evaluation due to the absence of attention pooling.

Model	Domain	Total Params	Base AUC	+MoLRE AUC	Trainable Params	MoLRE Params
<i>General 2D Vision Models</i>						
DINOv3-Base	Natural	86.6M	0.855 $\pm$ 0.083	0.902 $\pm$ 0.054	0.59M	0.28M
DINOv3-Large	Natural	305.2M	0.910 $\pm$ 0.052	0.912 $\pm$ 0.051	1.18M	0.37M
<i>Medical 2D Vision-Language Models</i>						
MedImageInsight	Medical	364.3M	0.863 $\pm$ 0.081	0.875 $\pm$ 0.081	1.42M	0.37M
MedGemma	Medical	431.1M	0.873 $\pm$ 0.084	0.917 $\pm$ 0.049	1.68M	0.41M
<i>Domain-Specific 3D Models</i>						
Pillar0-HeadCT	Head CT	113.9M	0.891 $\pm$ 0.068	0.893 $\pm$ 0.062	113.9M	0.41M
DeepCNTD	Head CT	6.9M	0.831 $\pm$ 0.091	–	1.08M	–

3.2 Multi-finding detection performance

Table 1 summarizes multi-label CT finding detection performance across 74 pathology classes. MoLRE consistently improves all compatible foundation models with less than 0.5% additional parameters; DeepCNTD [31] is excluded because it does not use attention pooling. Because our 2D MoLRE implementation routes slice-level embeddings before attention-based volume aggregation, models without this intermediate pooling interface were not directly compatible without architectural modification. Absolute AUC gains range from +0.2% to +4.7%, with MedGemma+MoLRE achieving the highest mean AUC of 0.917. Gains are largest for general-purpose and medical-general 2D models (DINOv3-Base: +4.7%; MedGemma: +4.4%) and smaller for larger-capacity or domain-specific models (DINOv3-Large: +0.2%; Pillar0-HeadCT: +0.2%). Together with the stable sensitivity results in Sect. 2.2, this suggests that MoLRE’s benefit depends more on domain shift, architecture, and pretraining strategy than on hyperparameter tuning.

The different behavior of MedImageInsight and MedGemma further illustrates this interaction. MedImageInsight is trained with a discriminative image-text objective, whereas MedGemma uses a generative vision-language pretraining strategy. The larger gain for MedGemma suggests that conditional low-rank specialization can recover task-specific discriminative structure that may not be fully organized for pure visual classification in generatively optimized features. MoLRE yields smaller gains for Pillar0-HeadCT, likely because routing is applied after volumetric pooling, where slice-level heterogeneity has already been compressed. This suggests a trade-off: 3D models provide strong volumetric inductive bias, while 2D slice-based pipelines may preserve finer spatial heterogeneity for conditional expert routing.

3.3 Stratified Detection and Per-Finding Performance

Figure 2 shows that MoLRE primarily improves performance by increasing the number of findings in the high-confidence regime ($AUC \geq 0.90$), rather than uniformly shifting all classes. For DINOv3-Base, the number of high-performing findings increases from 32 to 42; for MedGemma, it increases from 34 to 47. Larger or more specialized models show smaller stratified changes, consistent with their smaller overall MoLRE gains.

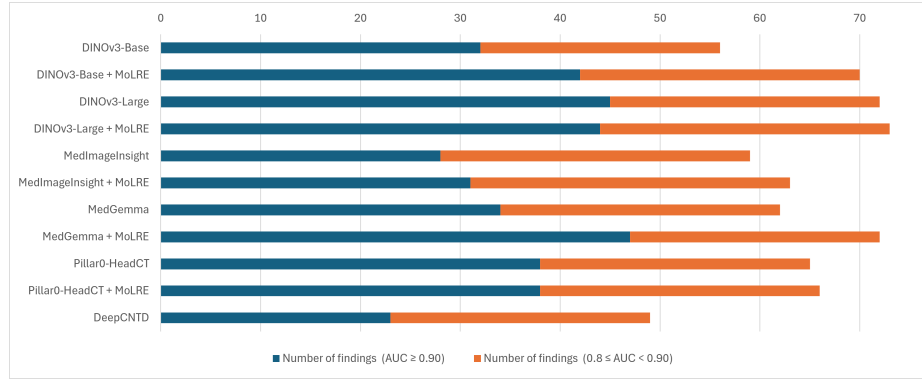


Fig. 2. Stratified multi-finding detection performance. Numbers of neurological findings achieving high-confidence performance ($AUC \geq 0.90$) and moderate performance ($0.8 \leq AUC < 0.9$) for each foundation model, with and without MoLRE.

Figure 3 shows that the largest per-finding gains occur for visually subtle, heterogeneous, or underrepresented findings. For DINOv3-Base, improvements are most apparent among ischemic, traumatic, mass-related, and vascular findings. For MedGemma, gains are more selective but include rare or complex entities with weaker baseline discrimination. In contrast, findings with high baseline performance, such as major hemorrhage or mass-effect patterns, show limited headroom. Overall, these results suggest that MoLRE functions as a targeted specialization mechanism, strengthening challenging or borderline findings rather than acting as a uniform performance booster.

3.4 Limitations and Future Work

This study has several limitations. First, although MoLRE was robust to reasonable choices of expert number and rank, detailed sensitivity analysis was performed primarily on DINOv3-Base; broader ablations across all backbones would further clarify model-specific behavior. Second, our results suggest that applying MoLRE after volumetric pooling may limit gains for 3D models, but targeted experiments inserting MoLRE at different spatial resolutions are needed

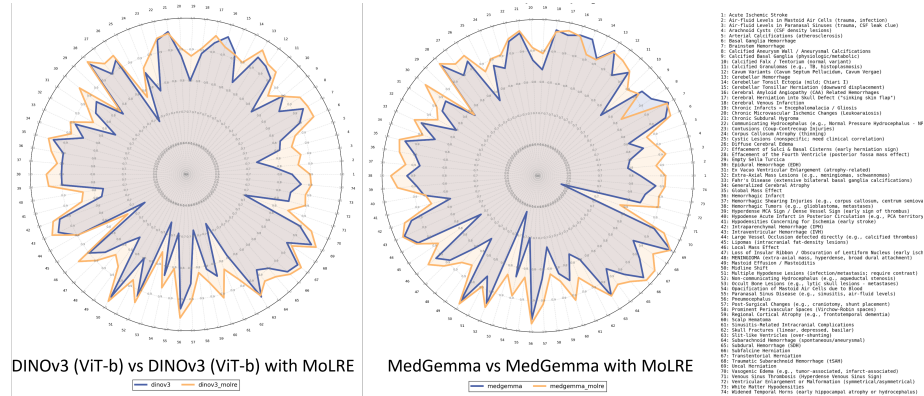


Fig. 3. Per-finding detection performance with and without MoLRE. Radar plots comparing baseline models and their MoLRE-enhanced counterparts for **left:** DINOv3-Base vs. DINOv3-Base+MoLRE and **right:** MedGemma vs. MedGemma+MoLRE across 74 neurological findings.

to verify this hypothesis. Third, while soft routing improves multi-label detection without explicit pathology supervision, further analysis of expert weights and routing patterns could help interpret what anatomical, acquisition-related or pathological variation each expert captures. Because $K = 1$ reduces MoLRE to a rank-matched single LoRA of equal capacity, our design cannot disentangle conditional routing from added capacity; future work should add uniform-routing and rank-matched single-LoRA baselines to isolate the routing contribution. Finally, prospective and workflow-level validation will be important before assessing clinical deployment impact.

4 Conclusion

In this work, we presented a comprehensive benchmark of Mixture of Low-Rank Experts across diverse medical imaging foundation models on large-scale, multi-label head CT analysis with 74 neurological findings. Our results demonstrate that MoLRE provides consistent benefits across models, with absolute AUC gains ranging from 0.2% to 4.7%, revealing that adaptation benefit depends on a complex interaction between pre-training domain and strategy, model architecture, and model size, rather than on model scale alone. By enabling conditional, parameter-efficient specialization, MoLRE allows general-purpose and medical-general foundation models to substantially improve performance, while task-specific or volumetric architectures exhibit diminishing but non-zero gains. Notably, MedGemma equipped with MoLRE achieves the best performance among the evaluated configurations, with an average AUC of 0.917. Because full fine-tuning was evaluated only for the 3D Pillar0-HeadCT backbone, we do not claim that lightweight expert routing matches or exceeds full fine-tuning in general; establishing this would require same-backbone comparisons (full fine-tuning versus

LoRA/MoLRE on an identical model), which we leave to future work. Beyond absolute performance, our stratified and per-finding analyses show that MoLRE primarily consolidates performance among challenging or borderline findings, elevating many into a high-confidence detection regime. Together, these findings provide new empirical insight into when and why conditional adaptation is effective, highlighting MoLRE as a practical and scalable strategy for specializing foundation models in complex clinical imaging tasks, particularly in resource-constrained deployment settings.

References

1. Agrawal, K.K., Liu, L., Lian, L., Nercessian, M., Harguindeguy, N., Wu, Y., Mikhael, P., Lin, G., Sequist, L.V., Fintelmann, F., et al.: Pillar-0: A new frontier for radiology foundation models. arXiv preprint arXiv:2511.17803 (2025)
2. Bicknell, B.T., Butler, D., Whalen, S., Ricks, J., Dixon, C.J., Clark, A.B., Spaedy, O., Skelton, A., Edupuganti, N., Dzubinski, L., et al.: ChatGPT-4 omni performance in usmle disciplines and clinical skills: Comparative analysis. *JMIR Medical Education* **10**(1), e63430 (2024)
3. Campanella, G., Hanna, M.G., Geneslaw, L., Mirafflor, A., Werneck Krauss Silva, V., Busam, K.J., Brogi, E., Reuter, V.E., Klimstra, D.S., Fuchs, T.J.: Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature medicine* **25**(8), 1301–1309 (2019)
4. Codella, N.C., Jin, Y., Jain, S., Gu, Y., Lee, H.H., Abacha, A.B., Santamaria-Pang, A., Guyman, W., Sangani, N., Zhang, S., et al.: Medimageinsight: An open-source embedding model for general domain medical imaging. arXiv preprint arXiv:2410.06542 (2024)
5. Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.M., Chen, W., et al.: Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature machine intelligence* **5**(3), 220–235 (2023)
6. Dou, S., Zhou, E., Liu, Y., et al.: LoRAMoE: Alleviating world knowledge forgetting in large language models via MoE-style plugin. In: *Proceedings of ACL*. pp. 1932–1945 (2024). <https://doi.org/10.18653/v1/2024.acl-long.106>
7. Fedus, W., Zoph, B., Shazeer, N.: Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* **23**(120), 1–39 (2022)
8. Flanders, A.E., Prevedello, L.M., Shih, G., et al.: Construction of a machine learning dataset through collaboration: the RSNA 2019 brain CT hemorrhage challenge. *Radiology: Artificial Intelligence* **2**(3), e190211 (2020)
9. Gupta, A., Dollar, P., Girshick, R.: Lvis: A dataset for large vocabulary instance segmentation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 5356–5364 (2019)
10. Hamamci, I.E., Er, S., Almas, F., Simsek, A.G., Esirgun, S.N., Dogan, I., Dasdelen, M.F., Wittmann, B., Simsar, E., Simsar, M., et al.: A foundation model utilizing chest CT volumes and radiology reports for supervised-level zero-shot detection of abnormalities. arXiv preprint arXiv:2403.17834 **5** (2024)
11. Hassanpour, S., Langlotz, C.P.: Information extraction from multi-institutional radiology reports. *Artificial intelligence in medicine* **66**, 29–39 (2016)
12. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al.: LoRA: Low-rank adaptation of large language models. *ICLR* **1**(2), 3 (2022)
13. Ilse, M., Tomczak, J., Welling, M.: Attention-based deep multiple instance learning. In: *International conference on machine learning*. pp. 2127–2136. PMLR (2018)
14. Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E.: Adaptive mixtures of local experts. *Neural computation* **3**(1), 79–87 (1991)
15. Jiao, L., Yang, J., Li, R., Liu, F., Liu, X., Chen, P., Guo, Y., Li, L., Shang, R., Ma, W.: Foundation model for medical imaging: A comprehensive review. *IEEE Transactions on Artificial Intelligence* (2025)
16. Khan, W., Leem, S., See, K.B., Wong, J.K., Zhang, S., Fang, R.: A comprehensive survey of foundation models in medicine. *IEEE Reviews in Biomedical Engineering* (2025)

17. Kiraly, F., Xu, Y., Chen, M., Kafle, K., Gemma-Team, et al.: MedGemma: Medical vision-language foundation models based on Gemma 3. arXiv preprint arXiv:2507.05201 (2025)
18. Lepikhin, D., Lee, H., Xu, Y., Chen, D., Firat, O., Huang, Y., Krikun, M., Shazeer, N., Chen, Z.: Gshard: Scaling giant models with conditional computation and automatic sharding. arXiv preprint arXiv:2006.16668 (2020)
19. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV) (2017)
20. Liu, H., Georgescu, B., Zhang, Y., Yoo, Y., Baumgartner, M., Gao, R., Wang, J., Zhao, G., Gibson, E., Comaniciu, D., et al.: Revisiting 2D foundation models for scalable 3d medical image classification. arXiv preprint arXiv:2512.12887 (2025)
21. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. International Conference on Learning Representations (ICLR) (2019)
22. Luo, T., Lei, J., Lei, F., Liu, W., He, S., Zhao, J., Liu, K.: MoELoRA: Contrastive learning guided mixture of experts on parameter-efficient fine-tuning for large language models. arXiv preprint arXiv:2402.12851 (2024). <https://doi.org/10.48550/arXiv.2402.12851>, <https://arxiv.org/abs/2402.12851>
23. Ma, J., Wang, B., Guo, X., Wang, J., Chen, Y., Zhang, Y., Wang, Y., Zhang, Q., Zhang, X., Tang, Y., et al.: Segment anything in medical images. *Nature Communications* **15**(1), 681 (2024). <https://doi.org/10.1038/s41467-024-44824-z>
24. Pons, E., Braun, L.M., Hunink, M.M., Kors, J.A.: Natural language processing in radiology: a systematic review. *Radiology* **279**(2), 329–343 (2016)
25. Ruder, S.: An overview of multi-task learning in deep neural networks. arXiv preprint arXiv:1706.05098 (2017)
26. Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G.: Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. arXiv preprint arXiv:1701.06538 (2017)
27. Siméoni, O., Vo, H.V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al.: Dinov3. arXiv preprint arXiv:2508.10104 (2025)
28. Titano, J.J., Badgeley, M., Schefflein, J., Pain, M., Su, A., Cai, M., Swinburne, N., Zech, J., Kim, J., Bederson, J., et al.: Automated deep-neural-network surveillance of cranial images for acute neurologic events. *Nature medicine* **24**(9), 1337–1341 (2018)
29. Wu, X., Huang, S., Wei, F.: Mixture of LoRA experts. In: International Conference on Learning Representations (2024)
30. Yang, S., Wu, X., Ge, S., Zheng, Z., Zhou, S.K., Xiao, L.: Radiology report generation with a learned knowledge base and multi-modal alignment. *Medical Image Analysis* **86**, 102798 (2023)
31. Yoo, Y., Georgescu, B., Zhang, Y., Grbic, S., Liu, H., Aldea, G.D., Re, T.J., Das, J., Ullaskrishnan, P., Eibenberger, E., et al.: A non-contrast head CT foundation model for comprehensive neuro-trauma triage. In: International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 3–13. Springer (2025)
32. Zhu, W., Huang, H., Tang, H., Musthyala, R., Yu, B., Chen, L., Vega, E., O'Donnell, T., Dehkharghani, S., Frontera, J.A., et al.: 3D foundation AI model for generalizable disease detection in head computed tomography. arXiv preprint arXiv:2502.02779 (2025)